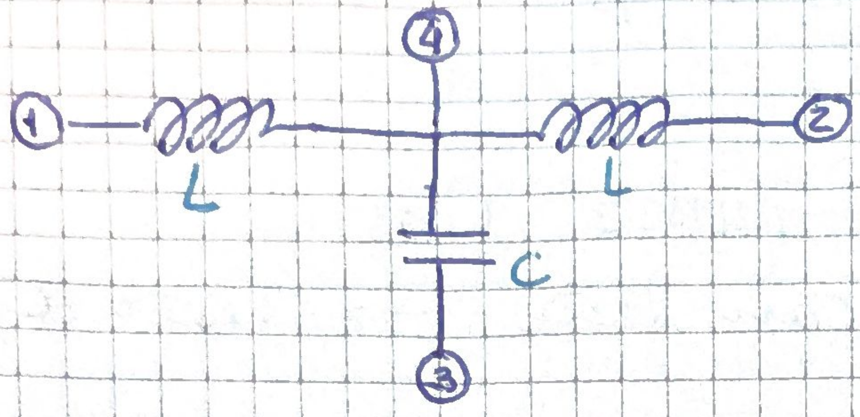
TS5 Lucas Gallero.

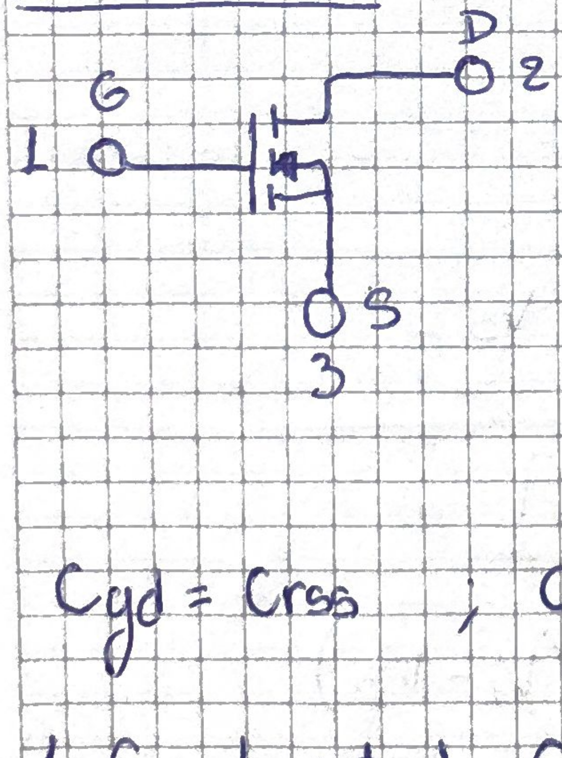
1er Circuito

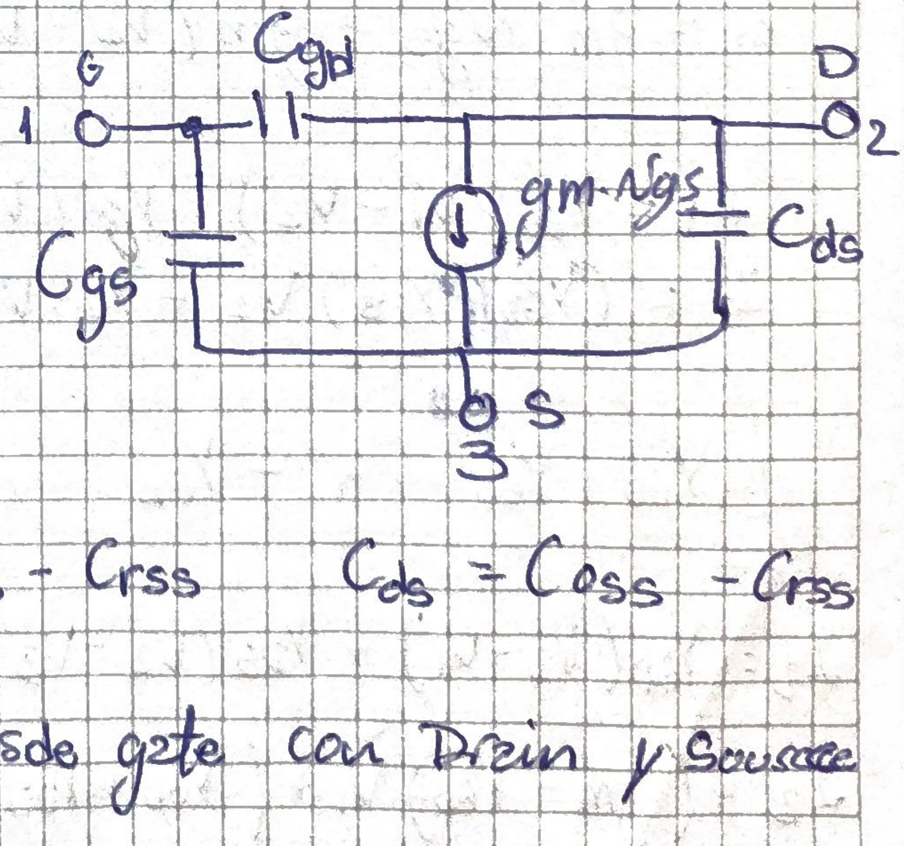


$Y_L = \frac{1}{SL}$; $Y_C = SC$

$$Y_{MAI} = \begin{bmatrix} Y_L & 0 & 0 & -Y_L \\ 0 & Y_L & 0 & -Y_L \\ 0 & 0 & Y_C & -Y_C \\ -Y_L & -Y_L & -Y_C & 2Y_L + Y_C \end{bmatrix}$$

2do Circuito





$C_{gd} = C_{rss}$; $C_{gs} = C_{iss} - C_{rss}$ $C_{ds} = C_{oss} - C_{rss}$

1. Cap. de entrada Ciss: Desde gate con Drain y Source puenteadar

$$C_{iss} = C_{gs} + C_{gd}$$

2. Cap. Salida C_{oss} : Desde Drain con Gate y Source puenteados

$$C_{oss} = C_{ds} + C_{gd}$$

3. Cap de Transferencia inversa C_{rss}

Es la que une Drain con Gate $\longrightarrow$ $C_{rss} = C_{gd}$.

$Y_{GD} = s\,C_{gd}$; $Y_{GS} = s\,C_{gs}$; $Y_{DS} = s$ ·

NODO 1 -> Gate NODO 2 -> Drain NODO 3 -> Source

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = Y_{MAT} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

$$i_m = g_m \cdot v_{gs} = g_m (V_1 - V_3)$$

$$I_1 = Y_{GD}(V_1 - V_2) + Y_{GS}(V_1 - V_3)$$
$$I_1 = (Y_{GD} + Y_{GS})V_1 - Y_{GD}V_2 - Y_{GS}V_3$$

$$\begin{bmatrix} Y_{GD} + Y_{GS} & -Y_{GD} & -Y_{GS} \end{bmatrix}$$

$$I_2 = Y_{GD}(V_2 - V_1) + Y_{DS}(V_2 - V_3) + g_m(V_1 - V_3)$$
$$I_2 = (g_m - Y_{GD})V_1 + (Y_{GD} + Y_{DS})V_2 + (-g_m - Y_{DS})V_3$$

$$\begin{bmatrix} g_m - Y_{GD} & Y_{GD} + Y_{DS} & -g_m - Y_{DS} \end{bmatrix}$$

$$I_3 = Y_{GS}(V_3 - V_1) + Y_{DS}(V_3 - V_2) - gm(V_1 - V_3)$$

$$I_3 = (-gm - Y_{GS})V_1 + (-Y_{DS})V_2 + (Y_{GS} + gm + Y_{DS})$$

$$\begin{bmatrix} -gm - Y_{GS} & -Y_{DS} & Y_{GS} + gm + Y_{DS} \end{bmatrix}$$

$$Y_{MA1} = \begin{bmatrix} Y_{GD} + Y_{GS} & -Y_{GD} & -Y_{GS} \\ gm - Y_{GD} & Y_{GD} + Y_{DS} & -gm - Y_{DS} \\ -gm - Y_{GS} & -Y_{DS} & gm + Y_{GS} + Y_{DS} \end{bmatrix}$$

Con valores de C_{rss}, C_{iss} y C_{oss}

$$Y_{MA1} = \begin{bmatrix} sC_{iss} & -sC_{rss} & -s(C_{iss} - C_{rss}) \\ gm - sC_{rss} & sC_{oss} & -gm - s(C_{oss} - C_{rss}) \\ -gm - s(C_{iss} - C_{rss}) & -s(C_{oss} - C_{rss}) & gm + s(C_{iss} + C_{oss} - 2C_{rss} \end{bmatrix}$$

Papel de fibra de caña de azúcar.

Punto 2

$$Y_{MH} = \begin{bmatrix} Y_L & 0 & 0 & -Y_L \\ 0 & Y_L & 0 & -Y_L \\ 0 & 0 & Y_C & -Y_C \\ -Y_L & -Y_L & -Y_C & Y_C+2Y_L \end{bmatrix}$$

$$\frac{V_{23}}{V_{13}}$$

$$Y_{(3)} = \begin{bmatrix} Y_L & 0 & -Y_L \\ 0 & Y_L & -Y_L \\ -Y_L & -Y_L & 2Y_L+Y_C \end{bmatrix}$$

$$\frac{V_{23}}{V_{13}} = \frac{C_{12}^{(3)}}{C_{11}^{(3)}}$$

$$C_{11}^{(3)} = \begin{vmatrix} Y_L & -Y_L \\ -Y_L & 2Y_L+Y_C \end{vmatrix} = Y_L(2Y_L+Y_C) - Y_L^2$$

$$C_{11}^{(3)} = 2Y_L^2 + Y_LY_C - Y_L^2 \Rightarrow \boxed{C_{11}^{(3)} = Y_L^2 + Y_LY_C}$$

$$C_{12}^{(3)} = (-1)^{1+2} \begin{vmatrix} 0 & -Y_L \\ -Y_L & 2Y_L+Y_C \end{vmatrix} = (-1)^3(-y_L^2).$$

$$\Delta = -y_L^2 \qquad \boxed{C_{12}^{(3)} = y_L^2}$$

$$\frac{V_{23}}{V_{13}} = \frac{Y_L \cancel{Z}}{\cancel{Y_L}(Y_L+Y_C)} = \frac{Y_L}{Y_L+Y_C} = \frac{1}{1+Y_C/Y_L}$$

$$\boxed{\frac{V_{23}}{V_{13}} = \frac{1}{1+S^2LC} = \frac{\frac{1}{LC}}{S^2+\frac{1}{LC}}}$$

$$\boxed{\frac{V_{21}}{V_{31}}} =$$

$$Y(1) = \begin{bmatrix} Y_L & 0 & -Y_L \\ 0 & Y_C & -Y_C \\ -Y_L & -Y_C & 2Y_L+Y_C \end{bmatrix}$$

$$\frac{V_{21}}{V_{31}} = \frac{C_{21}^{(1)}}{C_{22}^{(1)}}$$

$$C_{22}^{(1)} = \begin{vmatrix} Y_L & -Y_L \\ -Y_L & 2Y_L+Y_C \end{vmatrix} \rightarrow Y_L(Y_L+Y_C)$$

$$C_{21}^{(1)} = (-1)^{2+1}\begin{vmatrix} 0 & -Y_L \\ -Y_C & 2Y_L+Y_C \end{vmatrix} \rightarrow Y_L Y_C$$

$$\frac{V_{21}}{V_{31}} = \frac{\cancel{Y_L}\,Y_C}{\cancel{Y_L}(Y_L+Y_C)} = \frac{1}{\frac{Y_L}{Y_C}+1} = \frac{1}{\frac{1}{S^2LC}+1}$$

$$\boxed{\frac{V_{21}}{V_{31}} = \frac{S^2LC}{S^2LC+1}}$$

Papel de fibra de caña de azúcar.

$$\boxed{\frac{V_{12}}{V_{32}}} \qquad Y_{(2)} = \begin{bmatrix} Y_L & 0 & -Y_L \\ 0 & Y_C & -Y_C \\ -Y_L & -Y_C & 2Y_L + Y_C \end{bmatrix}$$

$$\frac{V_{12}}{V_{32}} = \frac{C_{21}^{(2)}}{C_{22}^{(2)}} \qquad C_{22}^{(2)} = Y_L(Y_L + Y_C)$$

$$C_{21}^{(2)} = (-1)^{2+1} \cdot Y_L Y_C$$

$$\frac{V_{12}}{V_{32}} = \frac{\cancel{Y_L} Y_C}{\cancel{Y_L}(Y_L + Y_C)} = \frac{Y_C}{Y_L + Y_C} = \frac{1}{\frac{Y_L}{Y_C} + 1}$$

$$\boxed{\frac{V_{12}}{V_{32}} = \frac{s^2 LC}{s^2 LC + 1}}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \qquad s^2 LC = -\omega^2 LC = -x^2$$

$$x = \frac{\omega}{\omega_0}$$

$$|T_1| = \frac{1}{|1 - x^2|}$$

$$|T_2| = |T_3| = \frac{|-x^2|}{|1 - x^2|} = \frac{x^2}{|1 - x^2|}$$

$$T_1 = \frac{1}{1 - x^2} \qquad T_2 = \frac{-x^2}{1 - x^2}$$

$$T_1 + T_2 = \frac{1}{1-x^2} + \frac{-x^2}{1-x^2} = \frac{1-x^2}{1-x^2} = 1.$$

Punto 3

a)
$$Y_{MAI} = \begin{bmatrix} s(C_{gD}+C_{gs}) & -sC_{gd} & -sC_{gs} \\ gm - sC_{gd} & s(C_{gd}+C_{ds}) & -gm - sC_{ds} \\ -gm - sC_{gs} & -sC_{ds} & gm + s(C_{gs}+C_{ds}) \end{bmatrix}$$

Y_{SC} $\rightarrow$ $V_S = V_3 = 0$

$v_1 = v_g - v_s = V_{13}$ $\qquad$ $v_2 = v_D - V_S = V_{23}$

$$Y_{SC} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix}$$

$$Y_{SC} = \begin{bmatrix} s(C_{gD}+C_{gs}) & -sC_{gd} \\ gm - sC_{gd} & s(C_{gD}+C_{ds}) \end{bmatrix}$$

$$\boxed{Y_{SC} = \begin{bmatrix} sC_{iss} & -sC_{rss} \\ gm - sC_{rss} & sC_{oss} \end{bmatrix}}$$

Papel de fibra de caña de azúcar.

$\boxed{Y_{DC}}$ $V_D = V_2 = 0$

$v_1 = v_g - V_d = V_{12}$ $v_2 = V_S - V_D = V_{32}$

$$Y_{DC} = \begin{bmatrix} Y_{11} & Y_{13} \\ Y_{31} & Y_{33} \end{bmatrix}$$

$$Y_{DC} = \begin{bmatrix} s(C_{gD} + C_{gs}) & -sC_{gs} \\ -g_m - s(C_{gs}) & g_m + s(C_{gs} + C_{ds}) \end{bmatrix}$$

$$\boxed{Y_{DC} = \begin{bmatrix} sC_{iss} & -s(C_{iss} - C_{rss}) \\ -g_m - s(C_{iss} - C_{rss}) & g_m + s(C_{iss} + C_{oss} - 2C_{rss}) \end{bmatrix}}$$

b) $A = -\dfrac{Y_{22}}{Y_{21}}$ $B = -\dfrac{1}{Y_{21}}$

$C = -\dfrac{\Delta Y}{Y_{21}}$ $D = -\dfrac{Y_{11}}{Y_{21}}$

$$\Delta Y = (sC_{iss})(sC_{oss}) - (-sC_{rss})(g_m - sC_{rss})$$

$$\Delta Y = s^2 C_{iss} C_{oss} + g_m sC_{rss} - s^2 C_{rss}^2$$

$$\Delta Y = s g_m C_{rss} + s^2 (C_{iss} C_{oss} - C_{rss}^2)$$

$$T_{sc} = \begin{bmatrix} -\dfrac{sC_{oss}}{g_m - sC_{rss}} & -\dfrac{1}{g_m - sC_{rss}} \\[2ex] -\dfrac{s g_m C_{rss} + s^2(C_{iss}C_{oss} - C_{rss}^2)}{g_m - sC_{rss}} & -\dfrac{sC_{iss}}{g_m - sC_{rss}} \end{bmatrix}$$